

SOC Monitoring Solution with **Wazuh & Suricata**



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PROBLEM ↗

- **The Cyber Threat**
- **The Core Operational Problem**
- **The Accessibility Problem**

Cyber Threat

Global By 2025



Cybercrime To Cost The World **\$10.5** Trillion Annually By 2025

Type of Threats

- Ransomware
- Phishing
- Nation & Organized Crime Hacking
- Malware Spying & Identity Theft
- Supply chained and deep web attack
- Infrastructure Attack

CYBERTHREAT LIVE MAP

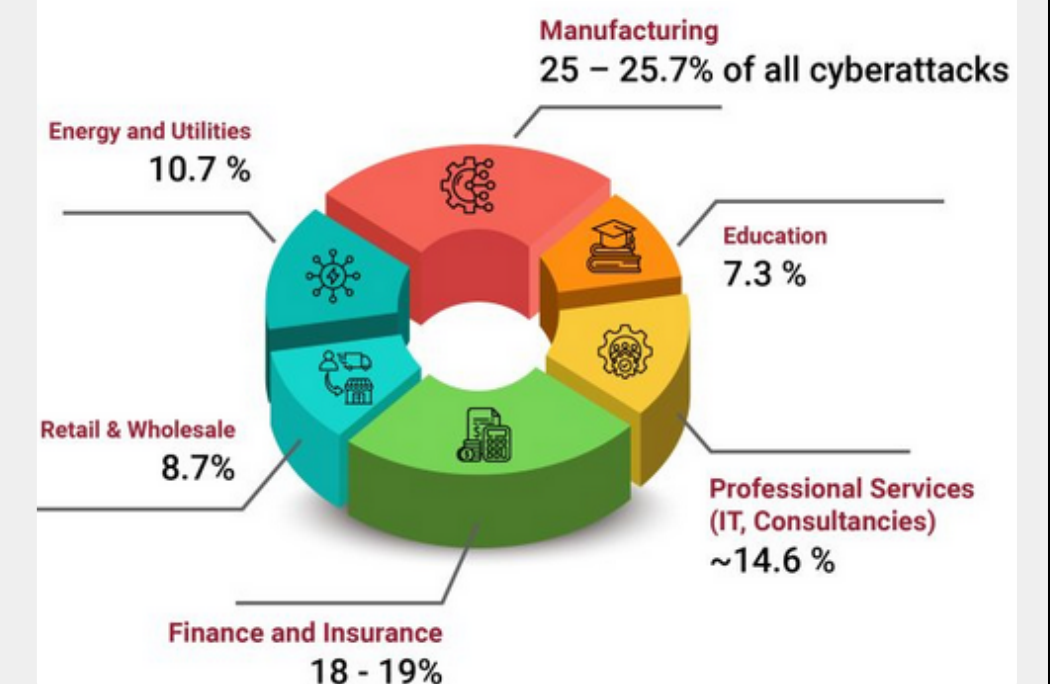


<https://cybermap.kaspersky.com>

6 Most Attacked Industries

- Manufacturing **25-27%**
- Education **7.3 %**
- Professional Service **14.6%**
- Finance and Insurance **18-19%**
- Retails and Wholesale **8.7%**
- Energy and Utilities **10.7%**

Top 6 most targeted U.S. industries in 2025



<https://www.ec-council.edu/top-industries-attacks>

Operational Problem



Stage of Attack	Suricata (Network)	Wazuh (Endpoint)	The SOC Reality
Reconnaissance	VISIBLE (Port Scans)	BLIND	You know you're being watched
Exploitation	VISIBLE (Exploit Code)	BLIND	You see the "bullet" in the air
Execution	BLIND	VISIBLE (Process Start)	You know the "bullet" hit the target
Persistence	BLIND	VISIBLE (Cronjob/Registry)	You see the attacker "unpacking bags"
Exfiltration	VISIBLE (Data Upload)	BLIND	You see the "stolen goods" leaving

Accessibility Problem

COMMERCIAL SIEM PLATFORM ARE EXPENSIVE AND LIMITED

Price Anually

- Splunk Enterprise Security : 600,000 \$
- IBM QRadar SIEM : 120,000 \$
- Microsoft Sentinel : 179,580 \$
- Exabeam Fusion : 196,000 \$

Affordability **NOT**

- Small Organization
- NGOs
- Educational Institution
- Government Agencies



Solution

AN OPEN-SOURCE SOC THAT ACTUALLY WORKS, INTRODUCING :



- **Focus:** Host-Based Security (HIDS & SIEM)
- **Monitoring :**
 - System logs and user authentication
 - File Integrity
 - Running processes and CPU spike
 - Vulnerabilities on the local OS
- **Typical Detection :**
 - Repeated failed login attempts (Brute Force)
 - A user executing a malicious script
 - Ransomware encrypting local files
 - Privilege escalation by an insider



- **Focus:** Network-Based Security (NIDS / NIPS)
- **Monitoring :**
 - Packet payloads and traffic flows
 - Protocol anomalies (HTTP, DNS, TLS)
 - Inbound/Outbound connections
 - Encrypted traffic patterns
- **Typical Detection :**
 - External port scanning (Reconnaissance)
 - SQL injections aimed at web servers
 - Malware communicating with a C2 Server
 - Large data exfiltration (Data Theft)

Defense in Depth : By combining Host-Based and Network-Based detection, the integrated solution provides full-spectrum visibility → closing the blind spots that attackers rely on.

→ Tools used

Wazuh

Purpose: Central Security Monitoring Platform

Key Functions

- Collect logs from endpoints
- Detect suspicious activities
- Monitor file integrity
- Generate security alerts

Monitored Data

- Linux system logs
- Windows event logs
- Authentication attempts
- System file changes

Output

- Alerts and dashboards for SOC analysts.

→ Tools used

Suricata

Purpose: Network Threat Detection

Capabilities

- Deep packet inspection
- Network traffic monitoring
- Protocol analysis
- Rule-based threat detection

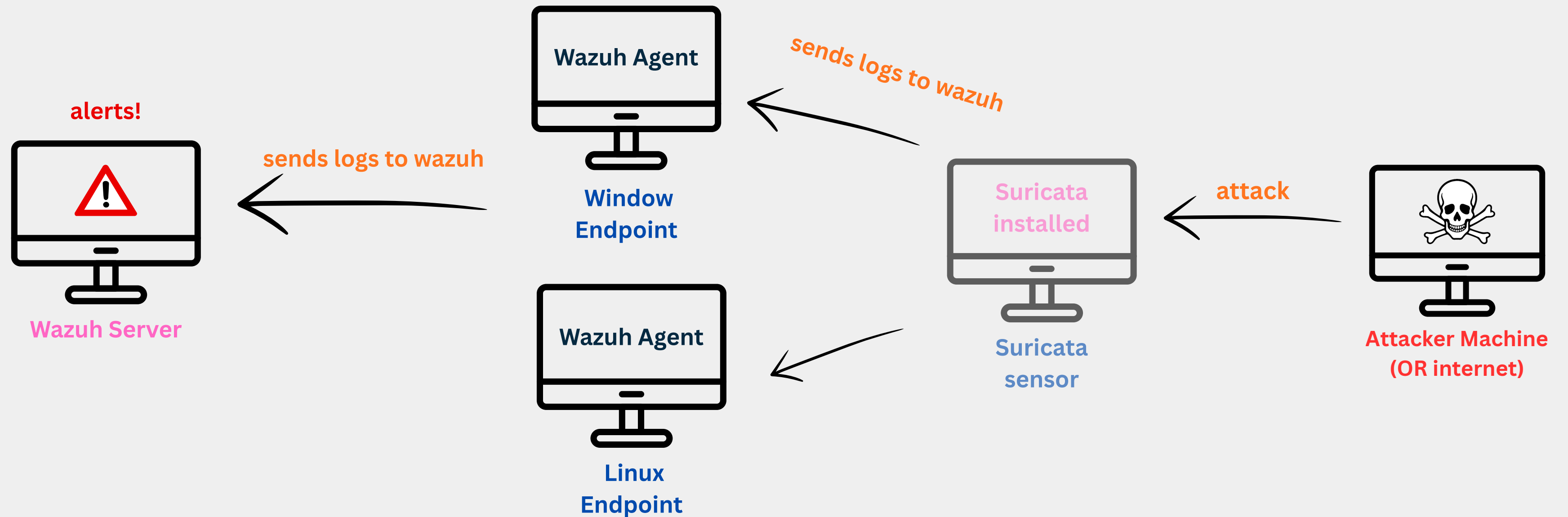
Detection Examples

- Port scanning
- Network reconnaissance
- SQL injection attempts
- DoS attacks

Integration

Suricata alerts are forwarded to Wazuh for centralized analysis.

➔ SOC Laboratory Environment



SOC Laboratory Environment

Five machines simulate a small SOC environment:

1. Wazuh Server
 - Manager + Indexer + Dashboard
2. Suricata Sensor
 - Network monitoring node
3. Linux Endpoint
 - Wazuh agent installed
4. Windows Endpoint
 - Wazuh agent installed
5. Attacker Machine
 - Performs simulated attacks:
 - Brute-force
 - Port scanning
 - Web attacks

Implementation Phase

Phase 1 : Infrastructure Foundation

- Installed Wazuh Manager, Indexer & Dashboard
- Configured network connectivity & baseline hardening

Phase 2 : Agent Deployment

- Deployed Wazuh agents on Linux & Windows endpoints
- Configured FIM, Auth Monitor, Sysmon & Malware Detection modules



Phase 3 : Network Monitoring Integration

- Installed & configured Suricata in IDS mode
- Integrated Emerging Threats ruleset
- Connected Suricata → eve.json → Wazuh for unified alerting

Phase : Testing & Validation

- Simulated 12 controlled attack scenarios
- Verified alert generation across all modules
- Documented SOC analyst response for each attack



What We Tested

#	Attack	Category	Why Selected
1	SSH Brute-Force Attack	Credential Attack	Most targeted remote access service in real environments
2	RDP Brute-Force Attack	Credential Attack	Common Windows attack vector, frequently exploited remotely
3	TCP SYN Port Scan	Reconnaissance	First step in almost every attack – host & service discovery
4	Service Version Enumeration	Reconnaissance	Banner grabbing reveals software versions to exploit
5	Unauthorized File Modification	File Integrity	Tests FIM detection – critical files changed by attacker
6	Web Shell Upload	Web Attack	Persistent backdoor giving attacker ongoing server access
7	Malicious Cron Job Entry	Persistence	Simulates attacker maintaining access via scheduled tasks
8	EICAR Test File	Malware Detection	Validates antivirus & file-based malware detection pipeline
9	Suspicious Encoded PowerShell	Evasion	Base64 obfuscation used to bypass security controls
10	Netcat C2 Channel (IRC)	Command & Control	Simulates post-exploitation attacker communication
11	SQL Injection via DVWA	Web Attack	OWASP Top 10 – most prevalent web application vulnerability
12	ICMP Flood & Ping Sweep	DoS / Recon	Validates volumetric & sweep detection at network level

Key Finding

What Worked Well:

- All 12 attacks detected and alerted in Wazuh dashboard
- Suricata + Wazuh integration provides both network and host-level visibility
- Open-source tools proven capable of enterprise-grade detection

Challenges We Discovered:

- EXTERNAL_NET must be set to any for LAN-based testing — by default Suricata only flags traffic coming from outside HOME_NET
- nmap -sn uses ARP on LAN instead of ICMP — bypasses ICMP-based detection rules entirely
- Wazuh has ~30 second indexing delay from Suricata alert to dashboard visibility

TEAM 5



Presentation



THANK YOU

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